

Unit 28: Indefinite Integrals and the Substitution Method

1. The idea, in plain words

The power rule for antiderivatives works great — as long as the “inside” of what you’re integrating is just plain x . But what happens with something like $\int (x^2 + 1)^3 (2x) dx$? That’s not a plain power of x — it’s a power of a whole expression, multiplied by something that looks suspiciously like the derivative of that expression. This is the exact pattern the chain rule creates when you differentiate — and substitution is simply the chain rule, run in reverse.

Picture it like unwrapping a package. The integral has a hidden “inner function” wrapped inside an “outer function,” and multiplied by the inner function’s own derivative — exactly the leftover crumbs the chain rule leaves behind. Substitution’s whole trick is to give that inner function a new, temporary name — u — so the messy expression turns into something the basic power rule (or basic trig rules) can handle directly.

The step-by-step process:

1. Spot the pattern: look for a function tucked inside another function, with the inner function’s derivative (or a constant multiple of it) sitting right there as a separate factor.
2. Let u equal the inner function.
3. Compute du — the derivative of u with respect to x , times dx .
4. Rewrite the entire integral in terms of u only — every single x needs to disappear, replaced by u ’s and du ’s. If you can’t fully clear out the x ’s, you may have picked the wrong u .
5. Integrate using the basic rules, now that it’s a simple expression in u .
6. Substitute back, replacing u with the original expression in x , and add $+C$.

For definite integrals, you have two options: either substitute back to x before plugging in the original bounds, or — often faster — convert the bounds themselves into u -values right when you substitute, and then never bother switching back to x at all.

2. Toolbox

The substitution method (indefinite integral):

$$\int f(g(x)) \cdot g'(x) dx \stackrel{u=g(x)}{=} \int f(u) du = F(u) + C = F(g(x)) + C$$

Changing the bounds for a definite integral (the shortcut — no need to substitute back):

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Common signals that a substitution is likely the right move:

- A power of a whole expression, multiplied by that expression’s derivative (or a constant multiple).
- A square root (or any fractional power) of an expression, multiplied by that expression’s derivative.
- A trig function applied to an expression other than plain x , multiplied by that expression’s derivative.

3. Common mistakes

- Choosing a u that doesn't actually simplify the integral. If, after substituting, you still have leftover x 's that won't cancel, try a different choice of u .
- Forgetting to convert dx into du properly. If $du = g'(x)dx$, and the constant in front doesn't match exactly, you may need to divide (or multiply) both sides by that constant to solve for the piece you actually need.
- Leaving some x 's in the integral after substitution. A fully correct substitution should result in an expression purely in terms of u — if any x remains, something's gone wrong.
- Forgetting to substitute back to x at the end (for indefinite integrals) — your final answer should generally be expressed in the original variable, unless you're doing a definite integral where you changed the bounds to u -values instead.
- Mixing up the two approaches for definite integrals — either change the bounds to u -values and finish entirely in u , or substitute back to x and use the original bounds. Don't accidentally use x -bounds with a u -expression, or vice versa.
- Forgetting the $+C$ on indefinite integrals.

4. Problem Set

Warm-up

1. $\int (x^2 + 1)^3(2x) dx$
2. $\int (3x + 1)^4 dx$
3. $\int 2x\sqrt{x^2 + 4} dx$
4. $\int (x^3 + 2)^2(3x^2) dx$
5. $\int (2x - 5)^6 dx$
6. $\int x(x^2 + 3)^5 dx$
7. $\int \sin(3x) dx$

Standard

8. $\int \frac{4x}{\sqrt{2x^2 + 1}} dx$
9. $\int x^2(x^3 + 1)^4 dx$
10. $\int \cos(2x + 1) dx$
11. $\int \frac{2x}{(x^2 + 1)^2} dx$
12. $\int \sec^2 x \tan x dx$
13. $\int (2x + 3)(x^2 + 3x + 1)^2 dx$
14. $\int 3x^2 \sin(x^3) dx$

Challenge

15. $\int x\sqrt{x+1} dx$ (this one needs an extra step — after substituting, you'll need to rewrite x itself in terms of u)
16. $\int_0^1 2x(x^2+1)^3 dx$ (change the bounds to u -values)
17. $\int_0^2 \frac{x}{\sqrt{x^2+9}} dx$ (change the bounds to u -values)
18. $\int \sin^3 x \cos x dx$
19. $\int x^3 \sqrt{x^4+5} dx$

Applied

20. A population's growth rate is modeled by $\frac{dP}{dt} = \frac{100t}{\sqrt{t^2+9}}$ people per year. Use substitution to find the general form of $P(t)$.
21. A company's marginal revenue is $MR(x) = 6x^2(x^3+10)^2$. Use substitution to find the general revenue function $R(x)$.
22. Water flows into a tank at a rate of $r(t) = \frac{8t}{(t^2+1)^2}$ gal/min. Use substitution to find the total water collected over $[0, 2]$ minutes.
23. An object's velocity is $v(t) = t \sin(t^2)$ m/s. Use substitution to find the exact distance traveled over $[0, \sqrt{\pi}]$.